Behavioral Alpha Rhythms Reveal Suppression of Predictable Distractors

Suyeon Kim & Oakyoon Cha Sogang University

Sogang University

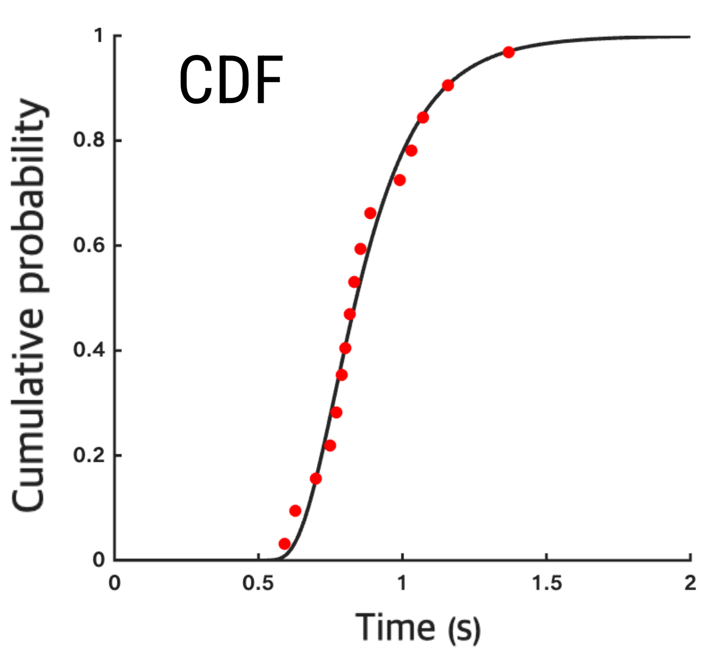
Decanting Conscious Awareness and Neural Timing DCANT Lab

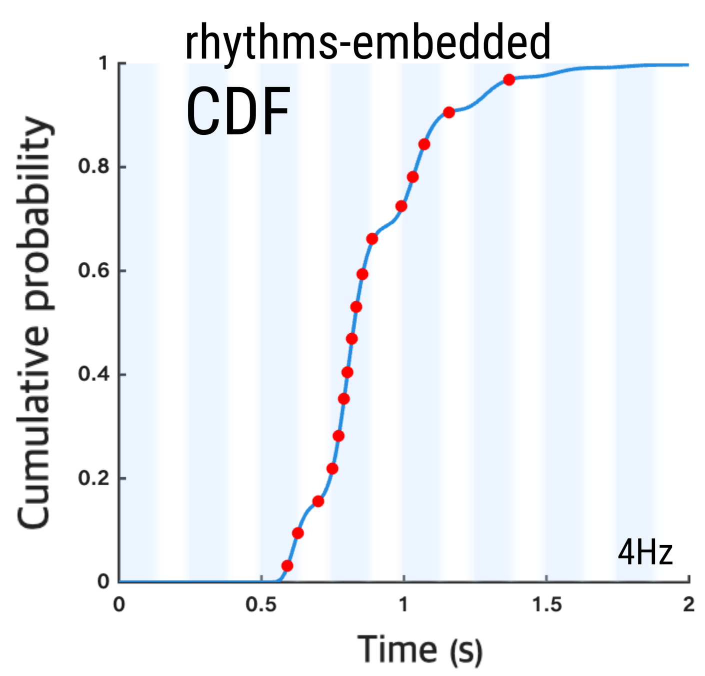
Background

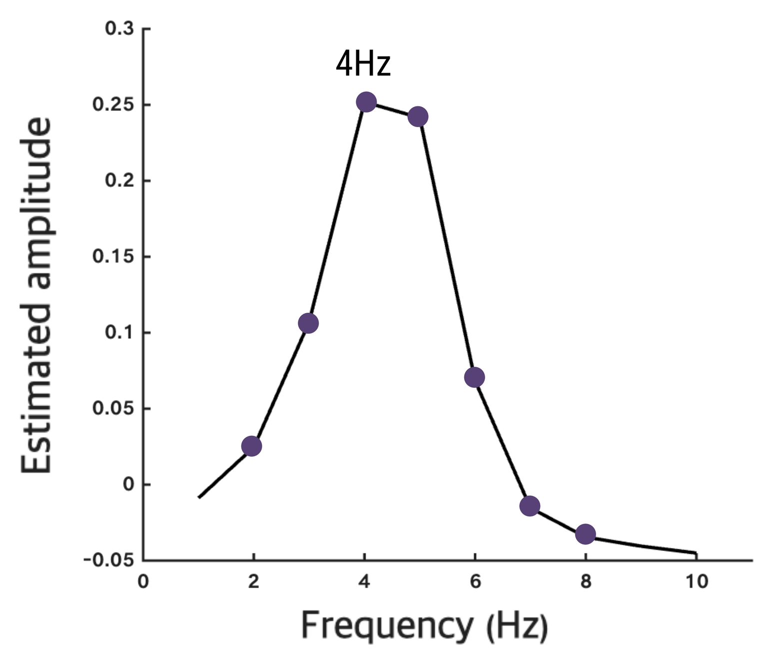
Alpha-band activity (EEG) reflects active suppression of irrelevant information, with alpha power increasing during distractor suppression (Jensen & Mazaheri, 2010).

PATS (Parametric Analysis of Temporal Structure)

PATS estimates a spectral power distribution by quantifying the strength of rhythmic fluctuations embedded in the CDF of duration data at each frequency (Cha & Blake, 2024).



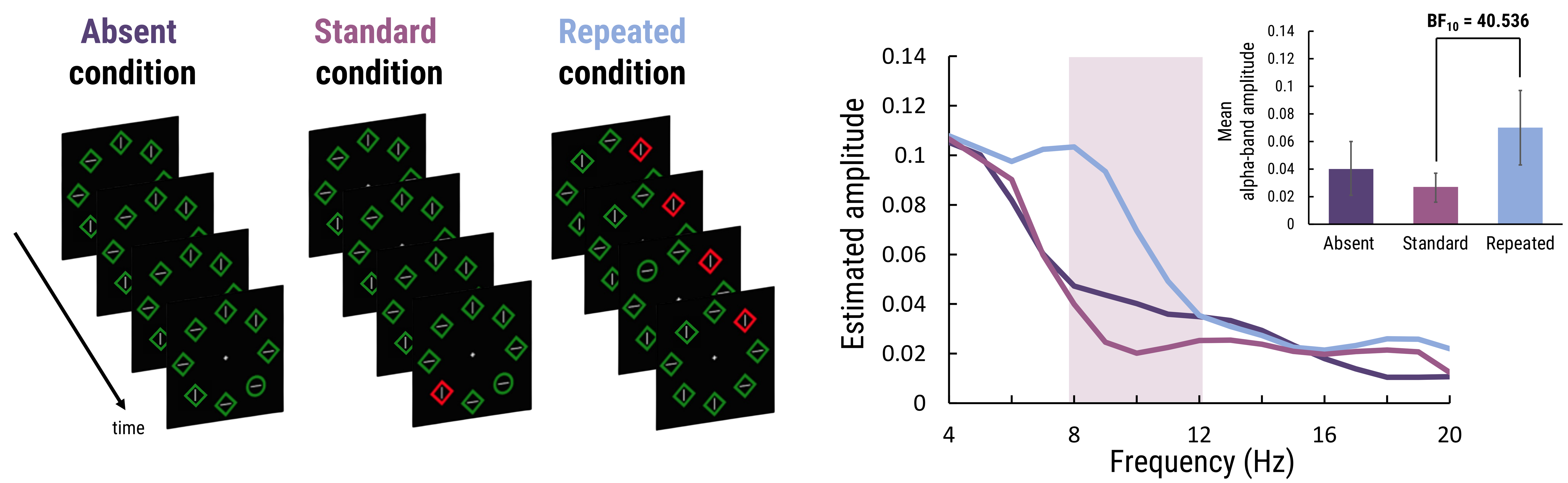




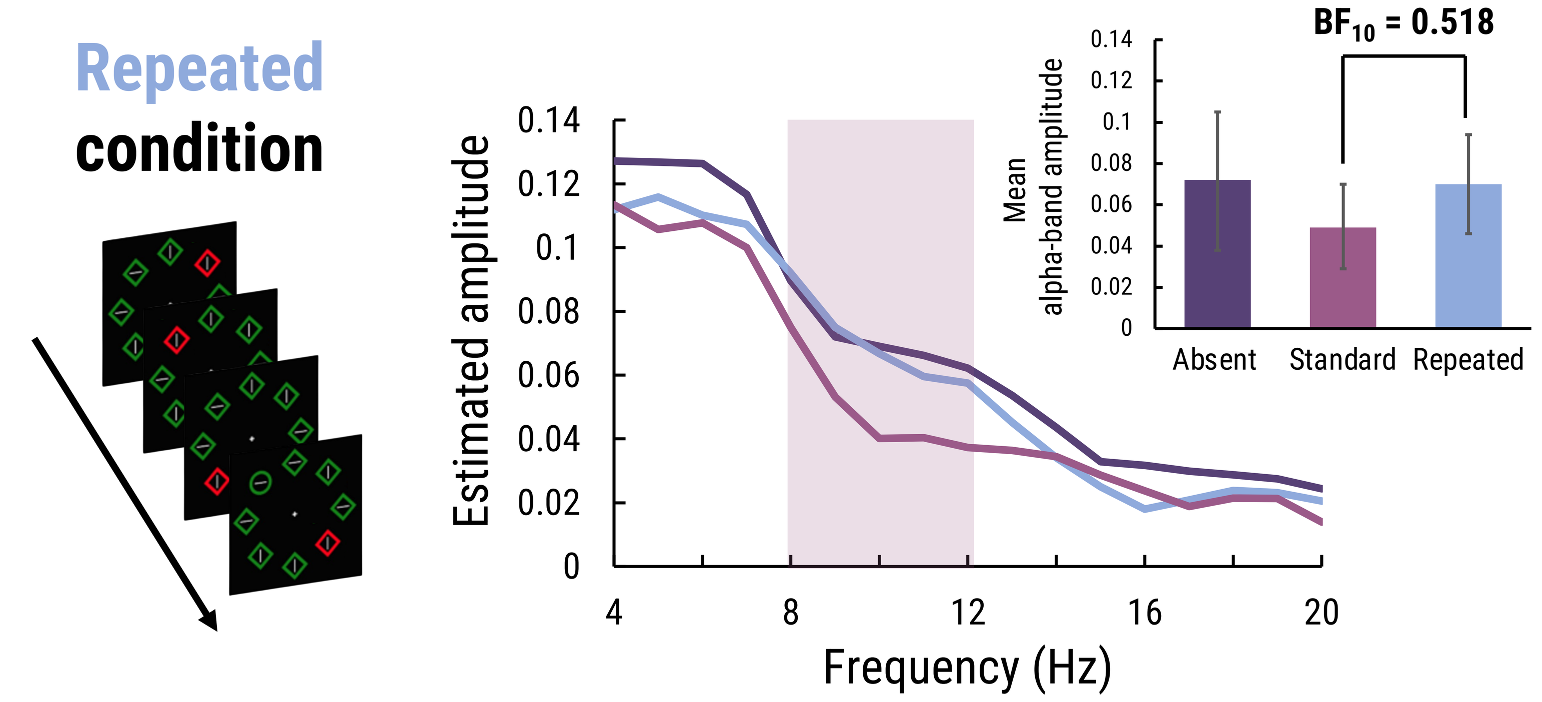
Reanalysis & Findings

from Betteto et al., 2025

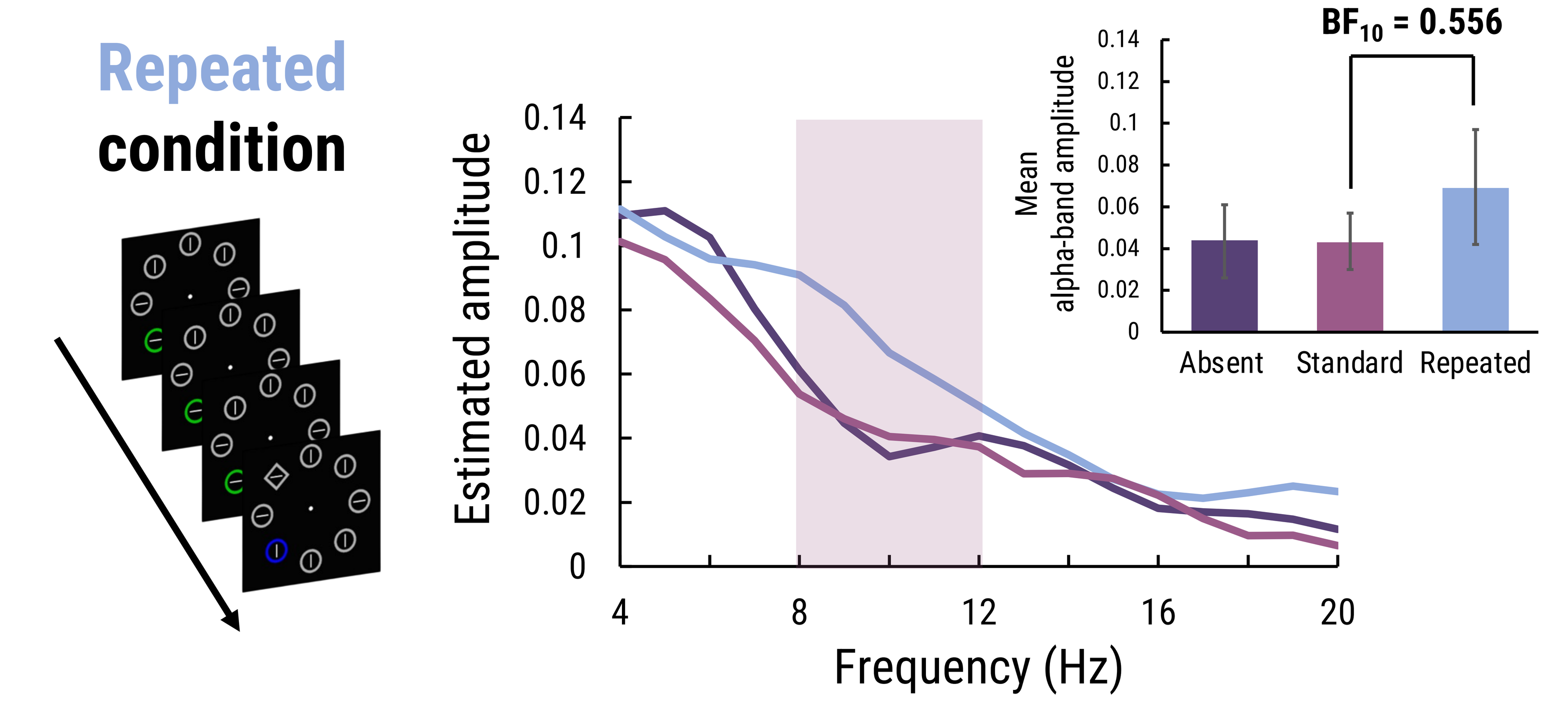
Experiment 1



Experiment 2



Experiment 3



Study Question

The present study examines whether alpha-band activity related to the suppression of predictable singletons manifests in response time data during visual search, using data from Betteto et al. (2025).

Conclusions

Alpha-band activity related to singleton suppression was captured in the response time distribution during visual search. Building on the Communication through Coherence framework (Fries, 2005), stronger alpha-band activity in the repeated condition (Exp. 1) may reflect phase resetting in preparation for the suppression of an upcoming distractor.

References

Betteto, L., De Tommaso, M., & Turatto, M. (2025). Rejecting an irrelevant singleton in the absence of a competing target. *Attention, Perception, & Psychophysics, 87*(4), 1083-1097.

Cha, O., & Blake, R. (2024). Procedure for extracting temporal structure embedded within psychophysical data. *Behavior Research Methods, 56*(6), 5482-5500.

Fries, P. (2005). A mechanism for cognitive dynamics: Neuronal communication through neuronal coherence. *Trends in Cognitive Sciences, 9*(10), 474-480.

Jensen, O., & Mazaheri, A. (2010). Shaping functional architecture by oscillatory alpha activity: Gating by inhibition. *Frontiers in Human Neuroscience, 4*, 186.

This work was supported by the National Research Foundation of Korea (NRF) grant funded by the Korean government (Ministry of Science and ICT) (No. RS-2023-00211668).